

Eclipse Setup

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[How to Install Banner Admin in IntelliJ](#)

The process of setting up Eclipse/Tomcat for Banner Admin and/or Banner Access Management (BAM) is exactly the same. The only differences between the two are:

- git repositories cloned (see *Required GIT Repos and Imports* below).
- Quick build directory differences. (See *Quick Builds* below).

Quick Guide

See this page [Banner Transformation - Clone Transformation Repository](#) for cloning details.

Some quick steps, but refer to this page for detail [quick_eclipse_setup](#)

[Banner Transformed Working with Eclipse and Your Administrative Application.pdf](#)

Set up [BPAPI](#) <https://ellucian.atlassian.net/wiki/x/HJbeFw>

Eclipse set up as of January 2023

[Foundations and Frames Eclipse Plugins Installation guide](#)

Maven Settings.xml file

Before you start you need to make sure your maven settings.xml file has the proper settings so you can download Morphis plugins and other files. Since both Banner Admin and BAM utilize Maven, we only need to do this once.

Create or open you .m2/settings.xml file.

Mac users the .m2 folder is located in your home or root folder (/Users/username/.m2).

Windows the .m2 folder is in your User folder.

Mar 22, 2023 Current settings for both oracle and PG

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <settings xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.1.0
http://maven.apache.org/xsd/settings-1.1.0.xsd"
```

```

3   xmlns="http://maven.apache.org/SETTINGS/1.1.0"
4   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
5   <profiles>
6     <profile>
7       <id>artifactory</id>
8       <repositories>
9         <repository>
10          <id>snapshots</id>
11          <url>https://artifactory.devops.ellucian.com/artifactory/admin-framework-
12          local/</url>
13        </repository>
14        <repository>
15          <id>releases</id>
16          <url>http://esmsvc02.ellucian.com:8181/nexus/content/groups/public</url>
17        </repository>
18      </repositories>
19    </profile>
20  </profiles>
21  <activeProfiles>
22    <activeProfile>artifactory</activeProfile>
23  </activeProfiles>
24  <servers>
25    <server>
26      <id>ellucian</id>
27      <username>ellucian</username>
28      <password>ellucian2014</password>
29    </server>
30  </servers>
31</settings>

```

This includes the framework internal repository, which represents the release version of the framework

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <settings xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.1.0
3    http://maven.apache.org/xsd/settings-1.1.0.xsd"
4    xmlns="http://maven.apache.org/SETTINGS/1.1.0"
5    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
6    <profiles>
7      <profile>
8        <id>artifactory</id>
9        <repositories>
10         <repository>
11           <id>snapshots</id>
12           <url>https://artifactory.devops.ellucian.com/artifactory/admin-framework-
13           local/</url>
14         </repository>
15         <repository>
16           <id>releases</id>
17           <url>https://artifactory.devops.ellucian.com/artifactory/admin-framework-
18           internal/</url>
19         </repository>
20         <repository>
21           <id>nexus</id>
22           <url>http://esmsvc02.ellucian.com:8181/nexus/content/groups/public</url>
23         </repository>
24       </repositories>
25     </profile>
26   </profiles>
27   <activeProfiles>
28     <activeProfile>artifactory</activeProfile>
29   </activeProfiles>
30   <servers>
31     <server>
32       <id>ellucian</id>
33       <username>ellucian</username>
34       <password>ellucian2014</password>
35     </server>
36   </servers>
37</settings>

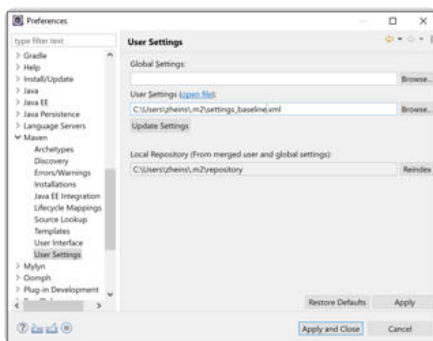
```

```

24     </repositories>
25   </profile>
26 </profiles>
27 <activeProfiles>
28   <activeProfile>artifactory</activeProfile>
29 </activeProfiles>
30 <servers>
31   <server>
32     <id>ellucian</id>
33     <username>ellucian</username>
34     <password>ellucian2014</password>
35   </server>
36 </servers>
37 </settings>

```

Since Postgres and Oracle development require different settings.xml files, it is possible to point to specific files for settings per workspace using similar settings as pictured below in Window > Preferences > Maven > User Settings and then pointing to a specific User Settings file.



Eclipse Versions

2019-12 is newest and Morphis supports but you must have newest version of foundations.
2.19.0

JDK

Install JDK 8 and use it for your page development.

New Banner Security Oracle and PG require JDK 11.

Morphis Foundation Plugins

You must be connected to VPN or Ellucian network to pull down plugins and also to do a maven update.

If you are installing Neon version of Eclipse install the following

<https://downloads.cdn.morphis-tech.com/eclipse/release/mars/>

Install the following for templates

<https://downloads.morphis-tech.com/eclipse/ellucian/release/4x>



For INTERNAL use only, here are the credentials you may need the first time you download the plugins

- User: ellucianuser
- Password: wW+E6?\$A

Make sure you accept all software updates. Foundations and Frames now utilizes a 3rd party software called Sapphire for storing preferences.

Current Version released to clients March 2019

Foundations Plugin 2.13.0

Frames 2.8.7

Templates Plugin 1.5

Version in test and with newer configurations

👍 Safe to start using now with Eclipse 2019-12

Foundations Plugin 2.19

Frames 2.8.8

Templates Plugin 1.7

Note: foundations 2.19 is required if you are using the 2019-12 and after of eclipse.

Global workspace settings

The following setup applies the Morphis plugin configuration to every workspace created. When these steps are followed, you will see, the foundation plugin, generator class preferences retained for every workspace and also it retains the preferences for the configuration section too.

This was tested in Mars 2

Locate the config.ini within eclipse/configuration folder and add this line to the bottom of the file.

For Mac users: Eclipse at least since Version: 2022-03 (4.23.0) on the Mac has been reworked to be a proper Mac application with everything included in the Eclipse.app package.

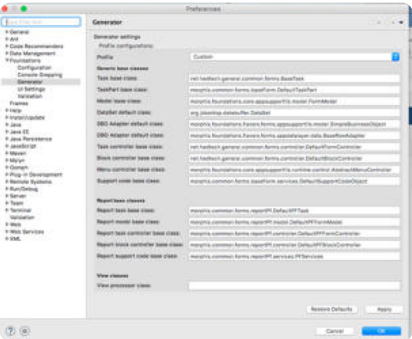
You can find the config.ini in /Applications/Eclipse.app/Contents/Eclipse/configuration/config.ini

1. Download the [morphis_eclipse.ini](#) file and place it in Applications/Eclipse.app/Contents/Eclipse/configuration
2. Add this line to the end of config.ini:
eclipse.pluginCustomization=<path>/morphis_eclipse.ini
For example: eclipse.pluginCustomization=/Users/edelaney/Downloads/morphis_eclipse.ini
3. Restart eclipse
4. Go to Eclipse/Settings/Foundations/Generator and confirm the Profile settings are filled in

Place the file in the location you specified above. When you restart the eclipse, you should see the configuration from the next section.

Foundation Generator

Access via the Window/Preferences menu.



Report base classes	
Report task base class	morphis.common.forms.reportPf.DefaultPFTask
Report model base class	morphis.common.forms.reportPf.model.DefaultPFFormModel
Report task controller base class	morphis.common.forms.reportPf.controller.DefaultPFFormController
Report block controller base class	morphis.common.forms.reportPf.controller.DefaultPFBBlockController

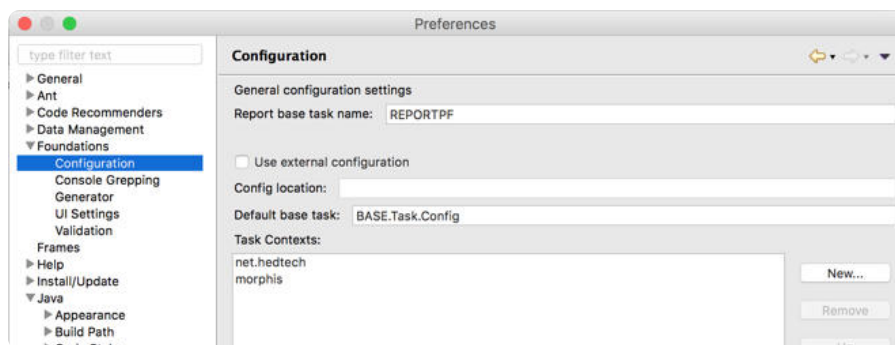
Report support code base class

morphis.common.forms.reportPf.services.PFSe
rvices

You can leave the View Classes blank.

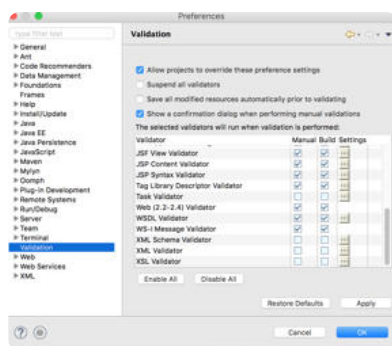
Foundations Configuration

Click on the foundations configuration and add net.hedtech and also change the default base task to BASE.Task.Config.



XML Validation

Make sure you uncheck the bottom 3 XML and XSL validator in the Eclipse / Preferences / Validation as shown here



Sizing

Setting the size and other related parameters are important so your app runs fast enough.

Basic sizing for JDK17 is:

```
1 -XX:MaxPermSize=512m -XX:PermSize=512m -Xms1g -Xmx4g -server -XX:+CMSClassUnloadingEnabled -
  XX:-UseGCOverheadLimit -
  DPROXY_SERVER_NAME=https://artifactory.devops.ellucian.com/artifactory/banner-aggregate/
```

Basic sizing for JDK17is:

```
1 -Xms1g -Xmx4g -server -
  DPROXY_SERVER_NAME=https://artifactory.devops.ellucian.com/artifactory/banner-aggregate/
2
3
```

Make sure you remove ' -Dhttp.proxyHost=www-proxy.sct.com' the proxy is no longer valid for our vpn; also be sure to remove the

```
PROXY_SERVER_NAME=http://esmsvc02.ellucian.com:8181/nexus/content/groups
/public -Dhttp.proxyPort=8080 as those are no longer necessary with the Nexus to A
rtifactory Migration
```

Eclipse Sizing

Follow instructions to increase the java heap size for your app by adding the xms, xmx to your eclipse.ini <http://crunchify.com/increase-eclipse-memory-size-to-avoid-oom-on-startup/>

Configuration Properties

Add the correct connection information to the
webapp.services/src/main/config/resources/config.properties

```
1 connection.host=localhost:1521/BAN83
2 connection.database=
```

For New Banner Security see page [Banner App Login Configuration](#). The login user is ban_admin_user for both New Banner Security Oracle and PG.

Turn on logging in that same properties file

```
1 log4j.logger.org.springframework.ws = TRACE
2 log4j.logger.org.springframework.ws.server.endpoint.interceptor = NONE
3 log4j.logger.org.springframework.beans.factory.xml = WARN
4 log4j.logger.org.springframework.context.support = WARN
5 log4j.logger.org.jgroups = ERROR
6 log4j.logger.morphis.foundations.core.appdatalayer = TRACE
7 log4j.logger.net.sf.jasperreports.engine.query.JRJdbcQueryExecuter = INFO
```

```

8 log4j.logger.morphis.foundations.core.jasper.JRRewindableJdbcQueryExecuter = INFO
9 log4j.logger.morphis.foundations.core.appsupportlib = TRACE
10 log4j.logger.net.hedtech.general.common.forms.services.BaseViewServices = INFO
11 log4j.logger.net.hedtech.general.common.security = DEBUG
12 log4j.logger.net.hedtech.general.common.services.localization = ERROR
13 log4j.logger.org.jasig.cas = ERROR
14 log4j.logger.morphis.frames.compiler = INFO
15
16

```

Thread message and Cache Warning

If you get lots of warning messages in the console for thread

```

1 Aug 18, 2022 1:49:10 PM org.apache.catalina.webresources.Cache getResource
2 WARNING: Unable to add the resource at [/WEB-INF/classes/

```

make sure you have the preload task listner commented out in the webapp.services web.xml files.

There are two web.xml files so make sure you get both.

```

1 <!-- Preload tasks -->
2 <!-- <listener>
3 <listener-
4 class>morphis.foundations.core.appsupportlib.configuration.TasksPreloader</listener-class>
5 </listener> -->

```

Caching

You can turn on or off the caching of pages in your tomcat context.xml file

Add to turn it on

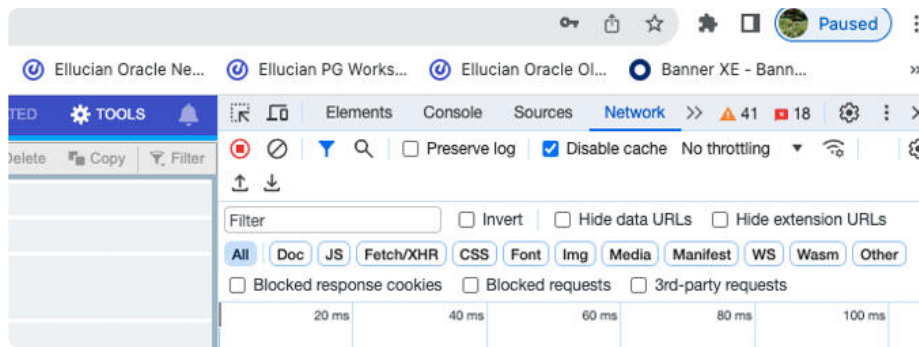
```
<Resources cachingAllowed="true" cacheMaxSize="102400" />
```

and turn it off

```
<Resources cachingAllowed="false" />
```

If your page does not reflect the changes you made to your view, try to turn off the caching in the developer tools.

Open the Network tab and click the Disable cache checkbox.



Other configuration changes are to the web.xml files

webapp.workspace/src/main/webapp/app/config/web.xml

comment out cache

```

1  <!--      <filter>
2           <filter-name>CacheFilter</filter-name>
3           <filter-class>com.samaxes.filter.CacheFilter</filter-class>
4           <init-param>
5               <param-name>expiration</param-name>
6               <param-value>86400</param-value>
7           </init-param>
8       </filter> -->
9
10      <filter>
11          <filter-name>NoCacheFilter</filter-name>
12          <filter-class>com.samaxes.filter.NoCacheFilter</filter-class>
13      </filter>
14
15      <!--      <filter-mapping>
16              <filter-name>CacheFilter</filter-name>
17              <url-pattern>/*</url-pattern>
18          </filter-mapping>
19
20          <filter-mapping>
21              <filter-name>NoCacheFilter</filter-name>
22              <url-pattern>/index.html</url-pattern>
23              <url-pattern>/config.xml</url-pattern>
24          </filter-mapping>      -->
25
26

```

webapp.servcies (build) /src/main/cnfig/WEB-INF/web.xml

Make sure you have these commented out

comment out listeners

```

1  <!-- <listener>
2           <listener-class>morphis.cache.ViewsCacheListener</listener-class>
3       </listener> -->
4
5       <!-- Spring sessions configuration -->
6
7  <!--
8       <listener>
9           <listener-
class>morphis.foundations.core.appsupportlib.configuration.session.SessionHelper</listener-

```

```

class>
10     </listener>
11
12     <context-param>
13         <param-name>contextConfigLocation</param-name>
14         <param-value>
15             /WEB-INF/spring-sessions.xml
16         </param-value>
17     </context-param>
18
19     <listener>
20         <listener-class>org.springframework.web.context.ContextLoaderListener</listener-
class>
21     </listener>
22
23     <filter>
24         <filter-name>springSessionRepositoryFilter</filter-name>
25         <filter-class>org.springframework.web.filter.DelegatingFilterProxy</filter-class>
26     </filter>
27     <filter-mapping>
28         <filter-name>springSessionRepositoryFilter</filter-name>
29         <url-pattern>/*</url-pattern>
30         <dispatcher>REQUEST</dispatcher>
31         <dispatcher>ERROR</dispatcher>
32     </filter-mapping>
33 -->

```

and the pretask loader

comment out cache

```

1     <!-- Uncomment to set number of threads used by TasksPreloader (default: 2/3 of total
processors available) -->
2     <!--
3     <context-param>
4         <param-name>TasksPreloaderThreads</param-name>
5         <param-value>1</param-value>
6     </context-param>
7     -->
8
9     <!-- Uncomment to set TasksPreloader thread priority ([1..10], default: 1) -->
10    <!--
11    <context-param>
12        <param-name>TasksPreloaderPriority</param-name>
13        <param-value>1</param-value>
14    </context-param>
15    -->
16
17    <!-- Preload tasks -->
18    <!-- <listener>
19        <listener-
class>morphis.foundations.core.appsupportlib.configuration.TasksPreloader</listener-class>
20    </listener> -->

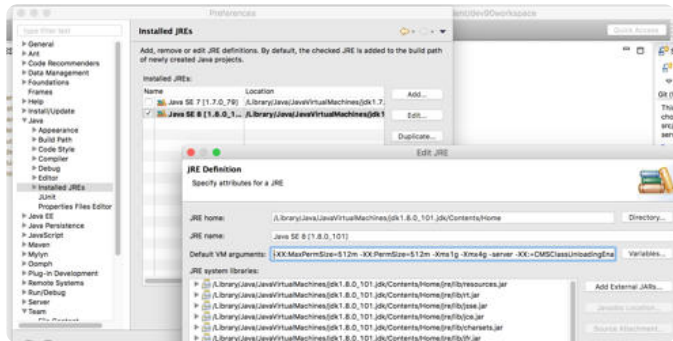
```

Environment Variables

In windows add the sizing to your JAVA_OPTS environment variable, and on MAC to the java_ops in your .bash_profile file.

Eclipse Heap Size

Open the eclipse properties / Java / Installed JREs. Ckeck your JRE and press the Edit Button.
Now enter the space variables in the VM arguments.



Switching between Oracle and Postgres

For Postgres :

1. in the webapp.services/src/main/resources/config.properties set the isSaas to Y for postgres and N for oracle

#for SaaS deployment, set this to Y for postgres and N for oracle

env.isSaas = Y

2. In the webapp.services/src/main/resources/appdatalayer.config.xml comment the imports lines 7-8 and 70-71

```
<!-- <import resource="appdatalayer.config_oracle.xml"/> -->
```

```
<import resource="appdatalayer.config_postgres.xml"/>
```

```
<!-- <property name="defaultDataBaseFactoryId" value="oracleDataBaseFactory"/> -->
```

```
<property name="defaultDataBaseFactoryId" value="postgreSQLDataBaseFactory"/>
```

3. in the config.properties set the data base

PG

```
1 connection.user           = ban_admin_user
2
3 connection.password       = u_pick_it
4
5 connection.host           = localhost:5432
6 connection.database       = ban2020
```

Oracle new Banner Security Oracle 19c

```
1 connection.user          = ban_admin_user
2
3 connection.password      = u_pick_it
4
5 connection.host          = localhost:5121/BAN83
6
7 connection.database      =
```

Oracle Old banner Security

```
1 onnection.user           = yourid
2
3 connection.password      = u_pick_it
4
5 connection.host          = localhost:5121/BAN83
6
7 connection.database      =
```

4. Switch the language in the config.properties

The following is for oracle

```
#####
# ENVIRONMENT
#####

#env.nls_date_format      = DD-MON-RRRR
#env.nls_canonical_lang   = en_US.UTF-8
##env.nls_canonical_lang  = AMERICAN
#env.nls_canonical_calendar = GREGORIAN
env.nls_sort              = BINARY
env.useDBInstanceNLSSettings = true

env.nls_date_format       = DD-MON-RRRR
#env.nls_canonical_lang   = en_US.UTF-8
env.nls_canonical_lang    = AMERICAN
env.nls_canonical_calendar = GREGORIAN
```

The following is for PG

```
#####
# ENVIRONMENT
```

```
#####
```

```
env.nls_date_format      = DD-MON-RRRR
env.nls_canonical_lang   = en_US.UTF-8
#env.nls_canonical_lang  = AMERICAN
env.nls_canonical_calendar = GREGORIAN
env.nls_sort             = BINARY
env.useDBInstanceNLSSettings = true

#env.nls_date_format     = DD-MON-RRRR
##env.nls_canonical_lang = en_US.UTF-8
#env.nls_canonical_lang  = AMERICAN
#env.nls_canonical_calendar = GREGORIAN
```

5. Set up the datasource configuration

In the build repo, make sure oracle (for dev_saas) has goktxml7.

dev_saas in banner admin has the new framework and requires changes to the goktxml package (goktxml7).

dev is the old framework and requires a different version of goktxml (goktxml).

The goktxml is defined in the build/webapp-services/src/main/resources/appdatalayer.config.xml

Make sure if you running bam or admin and working on the dev_saas branch, oracle, you have pulled the build repo.

Oracle Dev_saas New Framework

```
1 <bean id="appDataLayerSettings"
  class="morphis.foundations.core.appdatalayer.configuration.AppDataLayerSettings" init-
  method="initialize">
2     <property name="envVars">
3         <map>
4             <entry key="PKG_XML" value="goktxml7"/>
5             <entry key="useDBInstanceNLSSettings"
6 value="${env.useDBInstanceNLSSettings}"/>
7             <entry key="NLS_DATE_FORMAT" value="${env.nls_date_format}"/>
8             <entry key="NLS_CANONICAL_LANG"
9 value="${env.nls_canonical_lang}"/>
10            <entry key="NLS_CANONICAL_CALENDAR"
11 value="${env.nls_canonical_calendar}"/>
            <entry key="NLS_SORT" value="${env.nls_sort}"/>
        </map>
    </property>
```

Oracle dev Old Framework

```
1 <bean id="appDataLayerSettings"
  class="morphis.foundations.core.appdatalayer.configuration.AppDataLayerSettings" init-
```

```

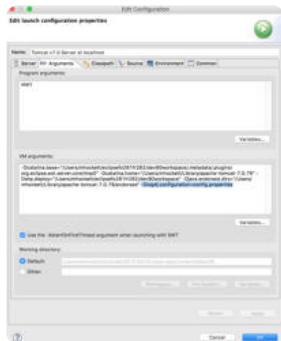
method="initialize">
2      <property name="envVars">
3        <map>
4          <entry key="PKG_XML" value="goktxml"/>
5          <entry key="useDBInstanceNLSSettings"
value="${env.useDBInstanceNLSSettings}"/>
6
7          <entry key="NLS_DATE_FORMAT">
8            <value>${env.nls_date_format}</value>
9          </entry>
10         <entry key="NLS_CANONICAL_LANG">
11           <value>${env.nls_canonical_lang}</value>
12         </entry>
13         <entry key="NLS_CANONICAL_CALENDAR">
14           <value>${env.nls_canonical_calendar}</value>
15         </entry>
16         <entry key="NLS_SORT">
17           <value>${env.nls_sort}</value>
18         </entry>
19       </map>
20     </property>

```

Run Time Configuration

Add the parameter for the configuration at the end of the VM arguments in the Run / Run Configuration.

`-Dlog4j.configuration=config.properties`



add the parameter `-Djava.io.tmpdir=/Users/mhockett/tempupload` to specify where upload files are saved.

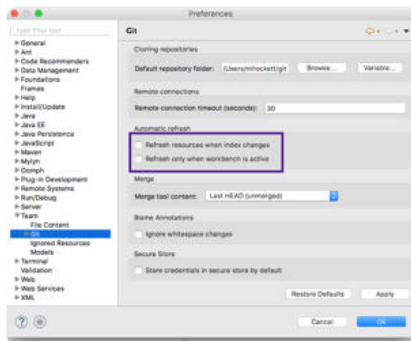
add the parameter `-Dlog4j.debug=true` to see other log messages from eclipse and the workspace

Team - GIT Indexing

Turn off automatic GIT indexing in your eclipse application.

Open Preferences / Team/ Git

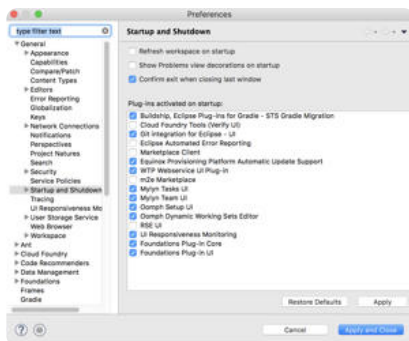
Uncheck the two checkboxes for automatic refresh.



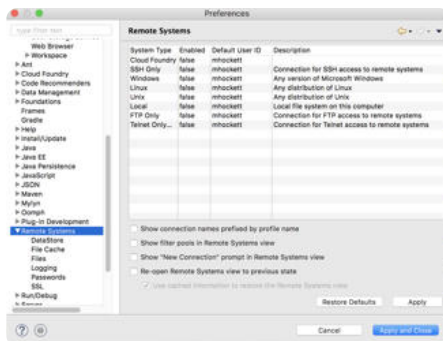
Other tips to speed up your eclipse

Remove some of the start up processes, like the shown here. Open preferences and type startup in the filter.

uncheck RSE UI and the others as they are not used for admin page development

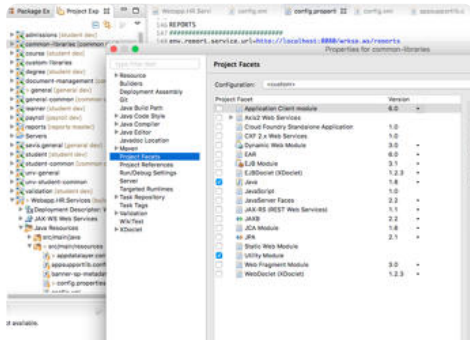


removing remote systems also helps

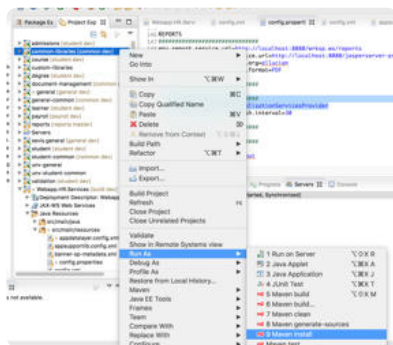


Troubleshoot

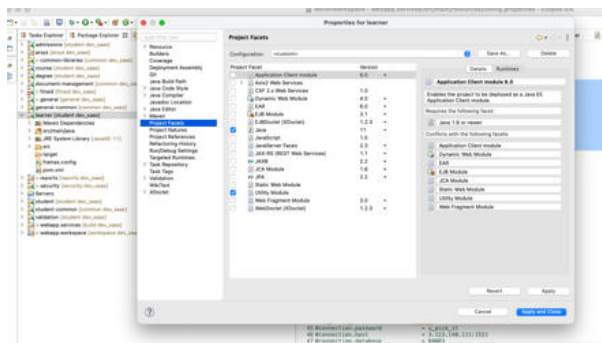
Make sure that these are set for your project. Right click on project, select properties, and check the two items as shown.



You can try to build common stand alone



Another tip is to make sure the project facet for each project has the right java and java utility right click on the project in the left panel, and select properties. Under Project Facet make sure the right java is there and clicked, and the utility module is clicked.



Speed up Tomcat startup

1. Open the two files:

```
1 webapp-services/src/main/config/WEB-INF/web.xml
2 webapp-services/src/main/webapp/WEB-INF/web.xml
```

2. Modify the top level xml attribute, add metadata-complete="true":

- a. Before:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app id="WebApp_Wrksp_WS_ID" version="2.5"
3     xmlns="http://java.sun.com/xml/ns/javaee"
4     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5     xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
6     http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd">
```

- b. After

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app id="WebApp_Wrksp_WS_ID" version="2.5"
3     xmlns="http://java.sun.com/xml/ns/javaee"
4     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5     xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
6     http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd"
7     metadata-complete="true">
```

Quick Build

1. Getting error during the maven update and can't determine the issue:

```
1 try these:
2 -- delete the morphis directory from the .m2/repository
3 cd into dev90/workspace/webapp-workspace # For Banner Admin
4 cd into bamdev90/workspace/webapp-workspace # For BAM
5 mvn clean install -U
6 -- if you get an error and you want to see it, try
7 mvn clean install -U -e
```

2. Quick clean instead of using Project / Clean

This is the same above but without the -U parameter

```
1 cd dev90/workspace/webapp-workspace # For Banner Admin
2 cd bamdev90/workspace/webapp-workspace # For BAM
3
4 mvn clean install
```

3. Maven Update at command line

```
1 cd dev90/build/net.hedtech.banner.hr # For Banner Admin
2 cd bamdev90/securitybuild/net.hedtech.banner.hr # For BAM
3 -- make sure pom.xml has all your projects and is correct
4 mvn package
```

BAM

[BAM Workspace extensibility](#)

Tomcat

you can run using tomcat 7 or 8, please be advised on bug with tomcat 8

<https://ellucian.force.com/clients/articles/FAQ/Cache-issues-when-using-Tomcat-8-x-in-an-Eclipse-development-environment>

Basically you may need to uncheck the serve modules without publishing if you are using tomcat 8, and then before each run you need to right click on the server and clean, and clean tomcat working directory and then publish.

Copy the jar files in the .m2/repositories/com/oracle/ojdbc7/12.1.10.1/ojdbc7-12.1.0.1.jar and .m2/repositories/com/oracle/ucp/11.2.0.4/ucp-11.2.0.4.jar to your tomcat/lib folder.

Required GIT Repos and Imports

Banner Admin

Start your development project by creating a main directory ban_admin_pages, with two sub directories dev90 and dev90workspace.

Clone your projects in to dev90, and open the ban_admin_ages/dev90workspace as your eclipse workspace

Copy the following clone shell script into your ban_admin_pages directory and execute.

- The following example clone shell script brings in the dev branch. Please review [Transformation GIT Branch Tracking#Overall](#) for current branch versions.

Sample clone shell script

```
1 # make home directory and two more for dev90 and eclipse workspace called dev90workspace
2 # mkdir -p ban_admin/dev90 ban_admin/dev90workspace
3 # cd ban_admin
4
5 # when you open eclipse the workspace is going to be
6 # ban_admin/dev90workspace and you'll import your projects from the
7 # ban_admin/dev90 directory
8 whereami=${PWD}
9 cd dev90
10
11 #required
12 git clone https://git.ellucian.com/scm/banadmpg/base.git --branch dev
13
14 git clone https://git.ellucian.com/scm/banadmpg/build.git --branch dev
15
16 git clone https://git.ellucian.com/scm/banadmpg/common.git --branch dev
```

```

17
18 git clone https://git.ellucian.com/scm/banadmpg/reports.git
19
20 #application repos
21 git clone https://git.ellucian.com/scm/banadmpg/general.git --branch dev
22
23 git clone https://git.ellucian.com/scm/banadmpg/finaid.git --branch dev
24
25 git clone https://git.ellucian.com/scm/banadmpg/studentaid.git --branch dev
26
27 git clone https://git.ellucian.com/scm/banadmpg/payroll.git --branch dev
28
29 git clone https://git.ellucian.com/scm/banadmpg/positioncontrol.git --branch dev
30
31 git clone https://git.ellucian.com/scm/banadmpg/extsol.git --branch dev
32
33 git clone https://git.ellucian.com/scm/banadmpg/bdr.git --branch dev
34
35 git clone https://git.ellucian.com/scm/banadmpg/student.git --branch dev
36
37 git clone https://git.ellucian.com/scm/banadmpg/alumni.git --branch dev
38
39 git clone https://git.ellucian.com/scm/banadmpg/finance.git --branch dev
40
41 git clone https://git.ellucian.com/scm/banadmpg/arsys.git --branch dev
42
43 # required
44 git clone https://git.ellucian.com/scm/banadmpg/workspace.git --branch dev
45
46 cd $whereami
47
48
49
50

```

Shell to update admin page DML

```

1 # make home directory and two more for dev90 and eclipse workspace
2 # mkdir ban_admin
3 # cd ban_admin && mkdir dev90 && mkdir dev90workspace
4 # when you open eclise the workspace is going to be ban_admin/dev90
5 # workspace and you'll import your projects from the dev90 directory
6
7 cd dev90
8 dev="${PWD}"
9
10 sh arsys/dev-scripts/tas_pages.shl "$dev/arsys"
11 sh finaid/dev-scripts/res_pages.shl "$dev/finaid"
12 sh general/dev-scripts/gen_pages.shl "$dev/general"
13 sh finance/dev-scripts/fin_pages.shl "$dev/finance"
14 sh payroll/dev-scripts/payroll_pages.shl "$dev/payroll"
15 sh positioncontrol/dev-scripts/posnctl_pages.shl "$dev/positioncontrol"
16 sh student/dev-scripts/stu_pages.shl "$dev/student"

```

The following projects are required to build your app, and must be imported into your workspace

common_libraries

document-management

general-common

reports

student-common

webapp.hr.workspace (workspace)

webapp.hr.services (build)

Banner Access Management

Start your development project by creating a main directory `ban_admin_pages`, with two sub directories `bamdev90` and `bamdev90workspace`.

Clone your projects in to `bamdev90`, and open the `ban_admin_ages/bamdev90workspace` as your eclipse workspace

Copy the following clone shell script into your `ban_admin_pages` directory and execute.

- The following example clone shell script brings in the dev branch. Please review [Transformation GIT Branch Tracking#Overall](#) for current branch versions.

Example clone for BAM

```
1  #!/bin/bash
2  declare ME="";
3  [[ ! -z "${USER}" ]] && ME="${USER}@";
4  git clone https://${ME}git.ellucian.com/scm/banadmpg/base.git -b dev
5  git clone https://${ME}git.ellucian.com/scm/banadmpg/common.git -b dev
6  git clone https://${ME}git.ellucian.com/scm/banadmpg/general.git -b dev
7  git clone https://${ME}git.ellucian.com/scm/banadmpg/security.git -b dev
8  git clone https://${ME}git.ellucian.com/scm/banadmpg/securitybuild.git -b dev
9  git clone https://${ME}git.ellucian.com/scm/banadmpg/reports.git -b master
10 git clone https://${ME}git.ellucian.com/scm/banadmpg/workspace.git -b dev
11
```

The following projects are required to build your app, and must be imported into your workspace

common-libraries

document-management

general

general-common

reports

student-common

[webapp.hr](#).workspace (workspace)

[webapp.hr](#).services (securitybuild)

security

Banner Admin and BAM directory structure

After creating the directory structure under *ban_admin_pages* and cloning Banner Admin and BAM, you should end up with:

```
1 ban_admin_pages
2 |-- bamdev90
3 |   |-- base
4 |   |-- common
5 |   |-- general
6 |   |-- reports
7 |   |-- security
8 |   |-- securitybuild
9 |   \-- workspace
10 |-- bamdev90workspace
11 |-- dev90
12 |   |-- base
13 |   |-- build
14 |   |-- common
15 |   |-- general
16 |   |-- reports
17 |   \-- workspace
18 \-- dev90workspace
```

BAM Build Configuration

Make sure the securityBuild pom file has the resource section within the build section commented out.

If you do not the config.properties used will be from the compiled classes and that will always have hrisusr and ellucian host.

Commenting out will ensure your config.properties in the src/resources folder is used.

```
1 <build>
2   <finalName>${profile.id}</finalName>
3   <plugins>
4     <plugin>
5       <artifactId>maven-war-plugin</artifactId>
6       <version>2.4</version>
7       <configuration>
8         <archive>
9           <addMavenDescriptor>>false</addMavenDescriptor>
10        </archive>
11        <warName>${profile.id}.ws</warName>
12        <webXml>src/main/config/WEB-INF/web.xml</webXml>
13        <webResources>
14          <resource>
15            <targetPath>WEB-INF/classes</targetPath>
16            <directory>src/main/resources</directory>
17          </resource>
18          <!-- <resource>
```

```

19         <targetPath>WEB-INF/classes/</targetPath>
20         <directory>src/main/config/WEB-INF/classes</directory>
21         <filtering>true</filtering>
22         <includes>
23             <include>config.properties</include>
24         </includes>
25     </resource> -->
26 </webResources>
27 <overlays>
28     <overlay>
29         <groupId>net.hedtech.banner.reports</groupId>
30         <artifactId>reports</artifactId>
31         <type>zip</type>
32         <classifier>resources</classifier>
33         <targetPath>/</targetPath>
34     </overlay>
35 </overlays>
36 <packagingExcludes>WEB-INF/lib/bcprov-jdk14*.jar</packagingExcludes>
37 </configuration>
38 </plugin>
39
40

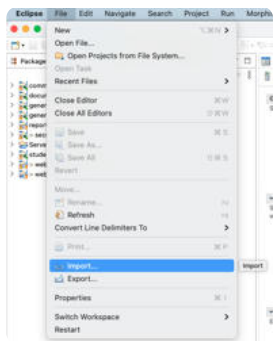
```

Importing projects into your workspace

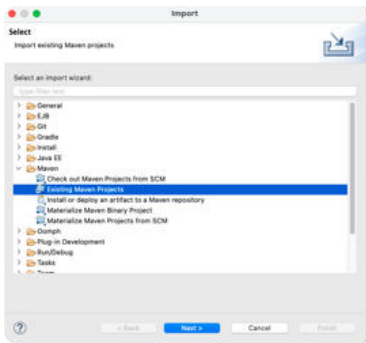
We now import as maven projects

[Troubleshooting Eclipse or Tomcat Failures with Banner 9x Administrative Projects](#)

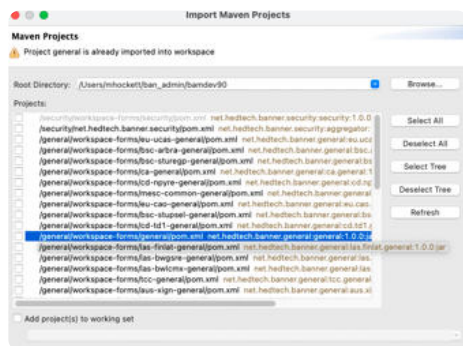
To import projects, from File select import



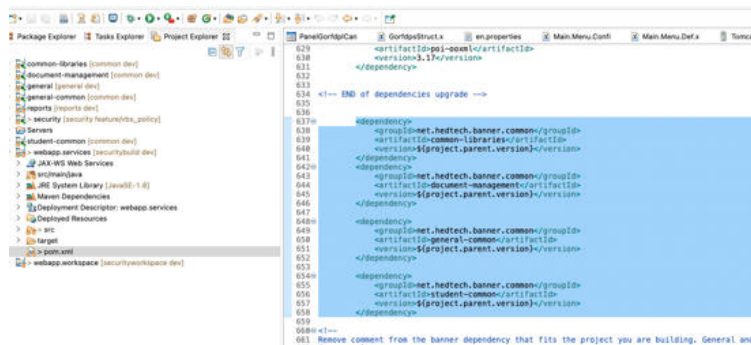
next select maven, existing maven projects



Set the root directory to be your dev90 or bamdev90 folder
select the pom.xml that are for workspace-forms, like the one selected below.



For each project selected, you need to make sure the entry in the webapp-services pom file is not commented out.



Creating a User ID

[Set Up Local Database to Run/Develop Transformation](#)

For new Banner Security [Create your test user in new Banner Security](#)

For PG Banner ID

Create pg user

```

1 DO $block$
2 DECLARE
3     sausr varchar(250) := 'saas_user';
4     p_pidm numeric;
5     p_pidm_inout numeric;
6     p_id_inout varchar(255);
7     p_rowid aws_oracle_ext.rowid;
8     p_rowid2 aws_oracle_ext.rowid;
9     tpid varchar(255);
10    l_sct_user varchar(30) := gb_common$f_sct_user();
11    c1 cursor (p_name spriden.spriden_id%type) for
12    select spriden_pidm
13    from   spriden
14    where  spriden_id = p_name
15    and    spriden_change_ind is null;
16 begin
17     p_pidm := null;
18     open c1 (sausr);
19     fetch c1 into p_pidm;
20     close c1;
21     raise notice 'PIDM is %', p_pidm::varchar;
22     if p_pidm is null then
23     call gb_identification$p_create(
24         p_id_inout      => sausr,
25         p_last_name     => sausr,
26         p_first_name    => sausr,
27         p_mi            => null::character varying,
28         p_change_ind    => null::character varying,
29         p_entity_ind    => 'P',
30         p_user          => l_sct_user,
31         p_origin        => null::character varying,
32         p_ntyp_code     => null::character varying,
33         p_data_origin   => 'BANNER'::character varying,
34         p_pidm_inout    => p_pidm,
35         p_rowid_out     => p_rowid
36     );
37     end if;
38     raise notice '%, %, %', sausr, p_pidm, p_rowid;
39     raise notice 'PIDM is %', p_pidm::varchar;
40     call gb_third_party_access$p_create(
41         p_pidm          => p_pidm,
42         p_pin_disabled_ind => 'N'::character varying,
43         p_usage_accept_ind => 'Y'::character varying,
44         p_user          => l_sct_user,
45         p_pin           => '111111'::character varying,
46         p_pin_exp_date  => null::timestamp without time zone,
47         p_question      => null::character varying,
48         p_response      => null::character varying,
49         p_insert_source => null::character varying,
50         p_ldap_user     => sausr,
51         p_data_origin   => 'BANNER'::character varying,
52         p_external_user_inout => tpid,
53         p_rowid_out     => p_rowid2);
54     insert into gobeacc (gobeacc_pidm, gobeacc_username, gobeacc_user_id,
55 gobeacc_activity_date)
56     SELECT p_pidm, sausr, current_user, aws_oracle_ext.sysdate()
57     WHERE NOT EXISTS (select 'x' from gobeacc where gobeacc_username = sausr);
58     insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
59 gurucsls_user_id)
60     select sausr, 'BAN_STUDENT_C', aws_oracle_ext.sysdate(), current_user
61     where not exists (
62         select 1
63         from gurucsls
64         where  gurucsls_userid = sausr
65               and gurucsls_class_code = 'BAN_STUDENT_C');
66 -- we use a single administrative user in PostgreSQL

```

```

65 --\connect - bansecl
66 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
67 select sausr, 'BAN_ARSYS_C', aws_oracle_ext.sysdate(), current_user
68 where not exists (
69     select 1
70     from gurucsls
71     where gurucsls_userid = sausr
72     and gurucsls_class_code = 'BAN_ARSYS_C');
73 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
74 select sausr, 'BAN_FINAID_C', aws_oracle_ext.sysdate(), current_user
75 where not exists (
76     select 1
77     from gurucsls
78     where gurucsls_userid = sausr
79     and gurucsls_class_code = 'BAN_FINAID_C');
80 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
81 select sausr, 'BAN_GENERAL_C', aws_oracle_ext.sysdate(), current_user
82 where not exists (
83     select 1
84     from gurucsls
85     where gurucsls_userid = sausr
86     and gurucsls_class_code = 'BAN_GENERAL_C');
87 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
88 select sausr, 'BAN_PAYROLL_C', aws_oracle_ext.sysdate(), current_user
89 where not exists (
90     select 1
91     from gurucsls
92     where gurucsls_userid = sausr
93     and gurucsls_class_code = 'BAN_PAYROLL_C');
94 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
95 select sausr, 'BAN_FINANCE_C', aws_oracle_ext.sysdate(), current_user
96 where not exists (
97     select 1
98     from gurucsls
99     where gurucsls_userid = sausr
100     and gurucsls_class_code = 'BAN_FINANCE_C');
101 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
102 select sausr, 'BAN_ADMIN_C', aws_oracle_ext.sysdate(), current_user
103 where not exists (
104     select 1
105     from gurucsls
106     where gurucsls_userid = sausr
107     and gurucsls_class_code = 'BAN_ADMIN_C');
108 insert into gurucsls ( gurucsls_userid, gurucsls_class_code, gurucsls_activity_date,
gurucsls_user_id)
109 select sausr, 'BAN_POSNCTL_C', aws_oracle_ext.sysdate(), current_user
110 where not exists (
111     select 1
112     from gurucsls
113     where gurucsls_userid = sausr
114     and gurucsls_class_code = 'BAN_POSNCTL_C');
115 p_pidm := null;
116 tpid := null;
117 end;
118 $block$;

```

Setting Trace to your Session

Launch tomcat with start_trace

Application

Application	URL	URL for Trace
Banner Admin	http://localhost:8080/wrksp/	http://localhost:8080/wrksp/?START_TRACE=YES
BAM	http://localhost:8080/wrksp/	http://localhost:8080/wrksp/?START_TRACE=YES